CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 2: **Electricity: Circuits and their Components** | Quick revision, one page

Components of a circuit

- Electric cell: converts chemical energy to electrical energy; long cap is positive (+), flat base is negative (−).
- Battery: two or more cells connected together; cells in series add their voltages (e.g. 1.5 V + 1.5 V = 3 V).
- Switch opens or closes the circuit; filament is the thin wire in a bulb that heats and glows.
- Electric circuit: a continuous path for current; closed circuit: current flows; open circuit: no current.

Conductors and insulators

- Conductors (copper, aluminium, iron, graphite, salt water) allow current to pass.
- Insulators (plastic, rubber, wood, glass, dry cloth) do not allow current to pass.
- Wires have plastic covering for safety; switch handles are made of insulators.

Series and parallel connection

- Series: components joined one after another; one path; if one fails all stop.
- Parallel: components joined across the same two points; many paths; others work if one fails.
- House wiring uses parallel connection.

Heating effect and safety

- Heating effect: a current passing through a wire heats it (electric heater, geyser, iron, toaster).
- Fuse: a wire with low melting point in series; melts when current is too large and breaks the circuit.
- MCB: automatic switch that trips on overloading or short circuit and can be reset.
- Short circuit: current takes a very low-resistance path and becomes very large; overloading: too many appliances on one line.

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